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BS-AI (Section B)

Programming for AI

Submitted To:

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Brain Tumor Classification Using Transfer Learning:

1. Introduction

Brain tumors are among the most critical neurological conditions requiring early and accurate diagnosis. Manual analysis of MRI scans is time-consuming and highly dependent on expert radiologists. This project presents a deep learning-based brain tumor classification system capable of classifying MRI images into four categories using transfer learning and explainable AI techniques.

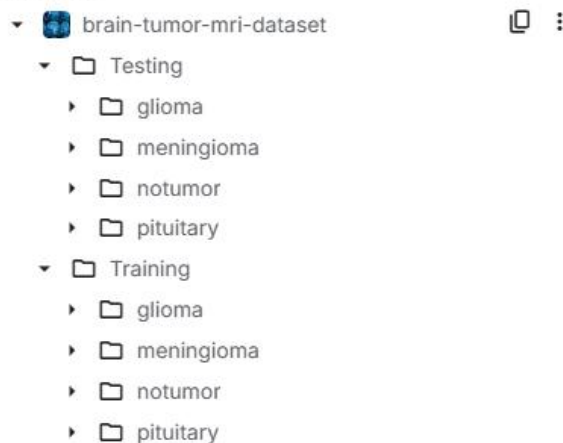
2. Objectives

- Classify brain MRI images into **four tumor categories**.
- Compare the performance of **two deep learning models**.
- Provide **visual explanation** of predictions using Grad-CAM
- Develop an easy-to-use **inference interface**.

3. Dataset Description

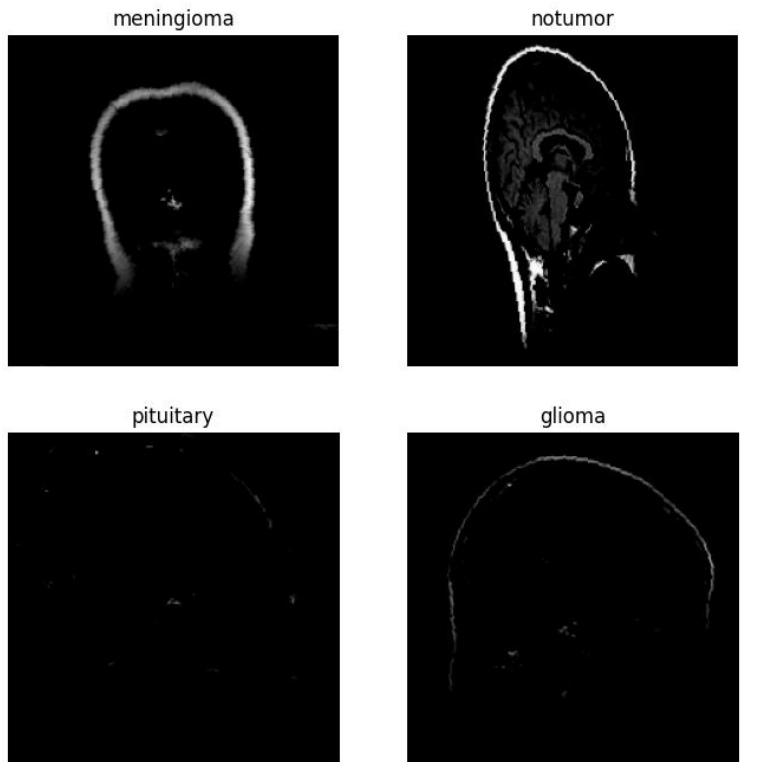
The Brain Tumor MRI Dataset was used for this project. The dataset is organized into training and testing directories, each containing four classes: Glioma, Meningioma, Pituitary, and No Tumor. Training dataset has atleast **6000 MRI** Images for training and testing dataset has at least **1500 MRI** Images.

DATASETS



4. Methodology

Image preprocessing steps included resizing all images to 224×224 pixels, normalization, and data augmentation techniques such as rotation and flipping to improve generalization.



5. Models Used

Two transfer learning models were used:

1. ResNet18
2. EfficientNet-B0

Both models were pre-trained on ImageNet and fine-tuned for four-class brain tumor classification.

6. Training Details

Optimizer: Adam

Loss Function: CrossEntropyLoss()

Epochs: 8

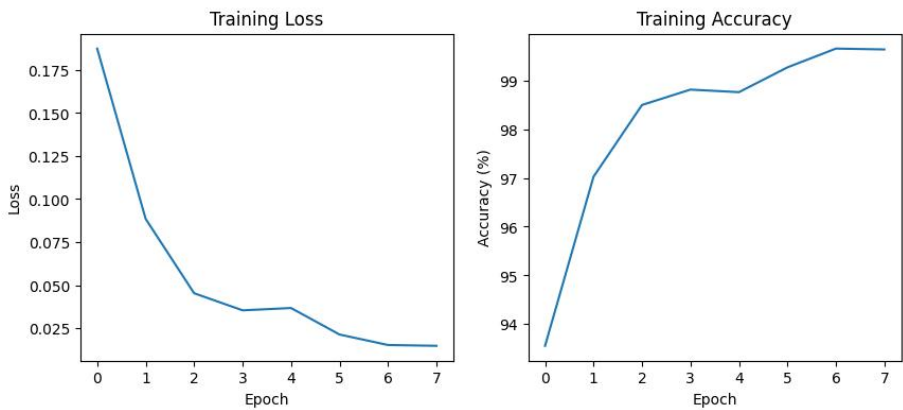
Batch Size: 32

7. Results and Evaluation

The models were evaluated using test accuracy, confusion matrices, and classification metrics.

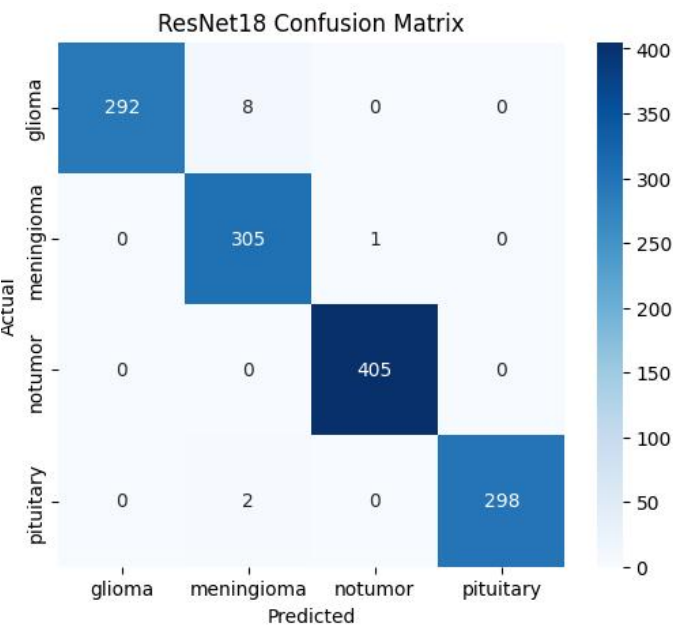
ResNet18:

Train Loss and Accuracy Graph:



Test Accuracy: 99.16%

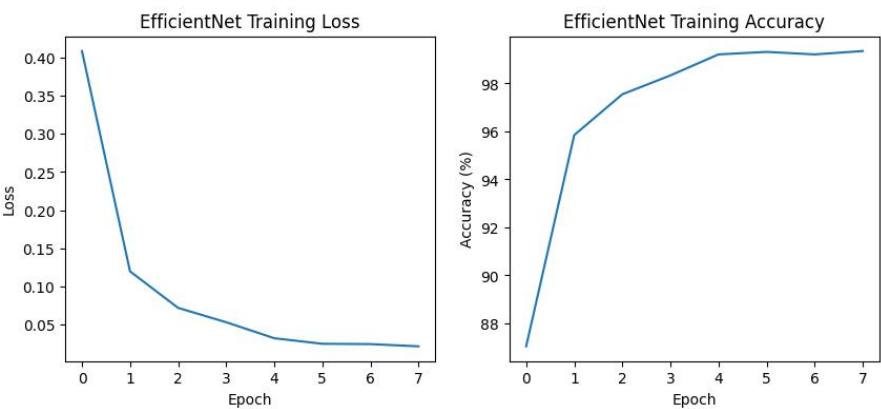
Confusion Matrix:



	precision	recall	f1-score	support
glioma	1.00	0.97	0.99	300
meningioma	0.97	1.00	0.98	306
notumor	1.00	1.00	1.00	405
pituitary	1.00	0.99	1.00	300

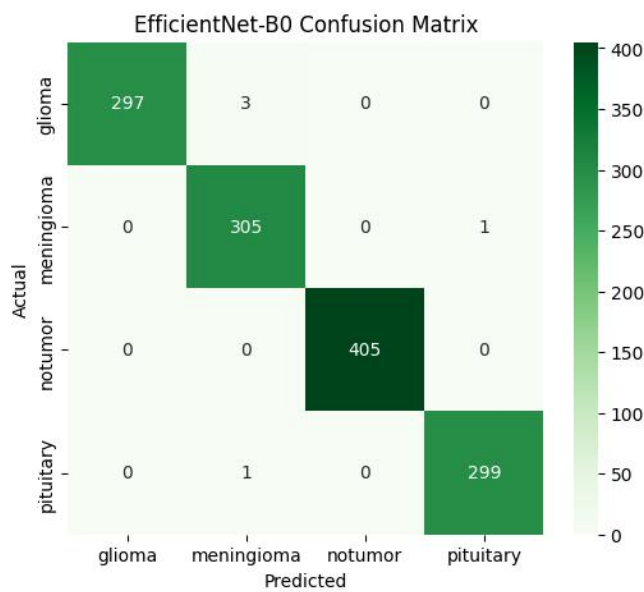
EfficientNet-B0:

Train Loss and Accuracy Graph:



Test Accuracy: 99.62%

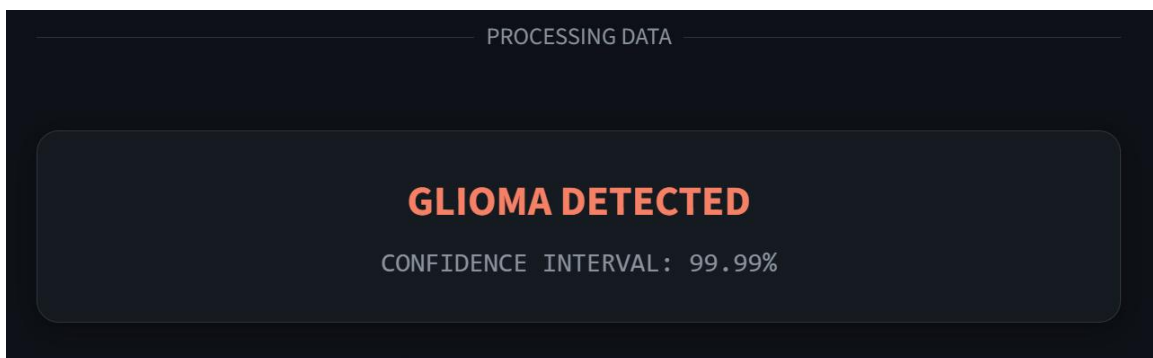
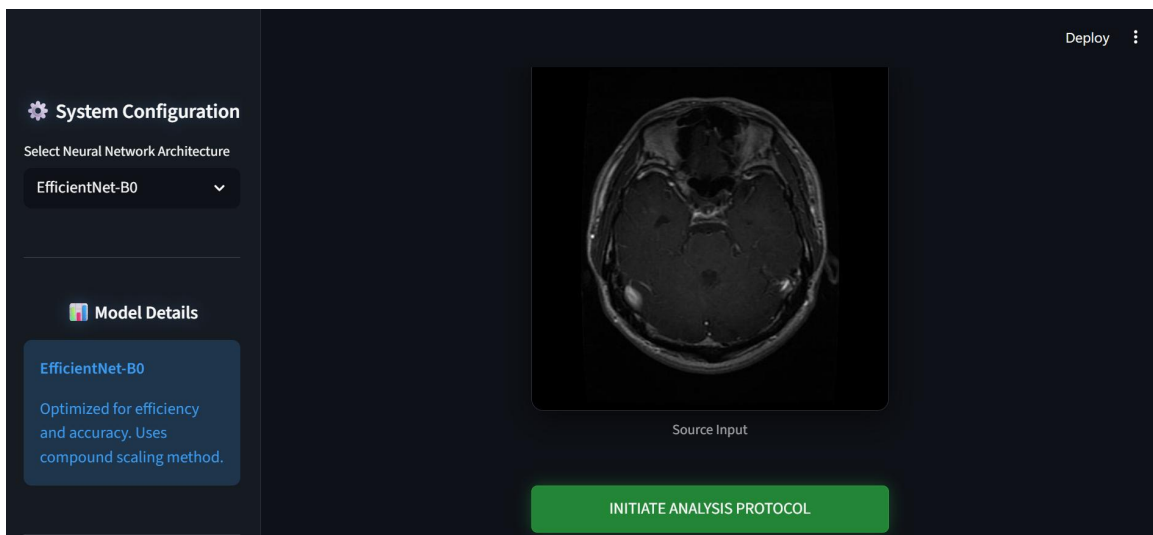
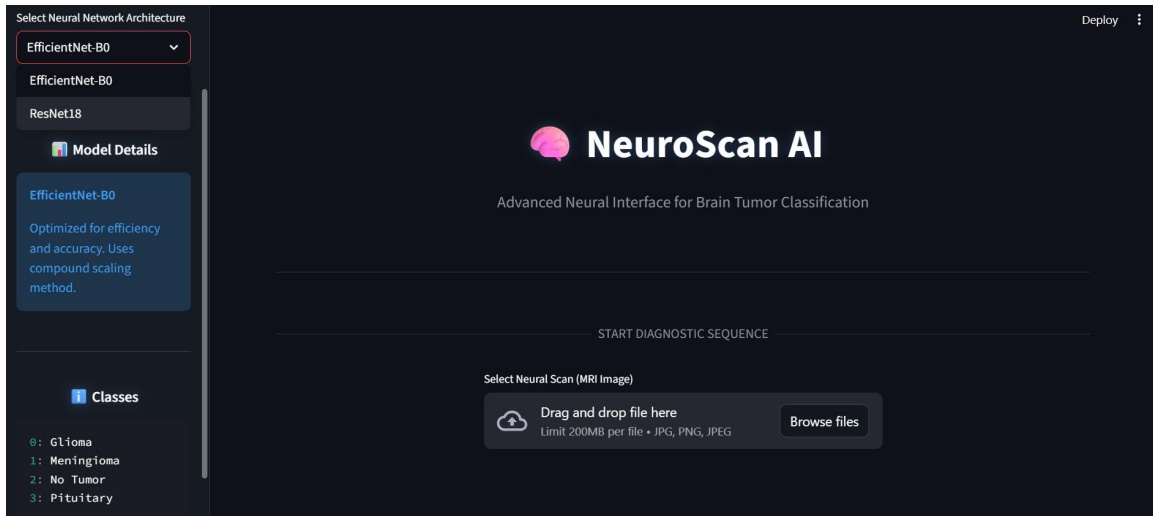
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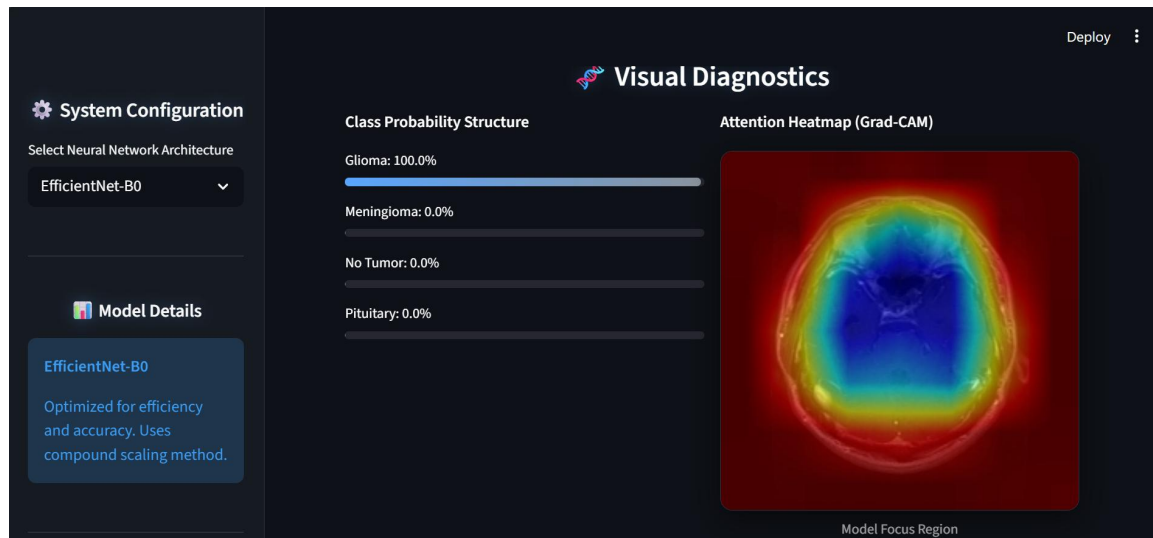
8. User Interface

A simple inference interface was developed in the Streamlit environment, allowing users to upload MRI images and view predictions along with confidence scores and Grad-CAM visualizations.



. Explainable AI (Grad-CAM)

Grad-CAM was applied to visualize the regions of MRI images that influenced the model's predictions. This improves transparency and trust in deep learning-based medical systems.



10. Conclusion

The EfficientNet-B0 model achieved the highest accuracy and demonstrated excellent generalization. The integration of explainable AI makes this system suitable as a decision-support tool.